

Subhrajit Prusty

SOFTWARE ENGINEER

Bengaluru, India

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Experience

Bolna

Bengaluru, India

LEAD BACKEND ENGINEER

January, 2026 - Present

- Building a voice AI orchestration platform in Python, handling tens of thousands of concurrent calls across Ecommerce, EdTech, HealthTech, BFSI, and Hospitality
- Re-architected post-call processing into a phased, idempotent pipeline with atomic claim-and-checkpoint semantics, plus reconciliation jobs that recover calls stranded mid-settlement (13k+ in one incident) and bounded slow provider-API retries that were pinning worker slots; hardened call-initiation paths against multi-pod race conditions — eliminated double-billing and lost-work bugs and enabled horizontal scaling without duplicate execution
- Cut per-pod worker memory by ~95% (from ~15GB to under 1GB) through WebSocket lifecycle cleanup, and moved ws-server to a gunicorn preload-fork model so workers share a single preloaded copy of the application via copy-on-write — saving a further 1–2 GiB per replica
- Drove custom-metric and queue-depth horizontal pod autoscaling on real application signals instead of CPU alone, tuned unstable scaling signals (e.g. replaced an api-server p95-latency HPA with CPU), and right-sized requests/limits with topology-spread constraints across the worker fleet to balance node utilization and remove OOM and bin-packing hotspots
- Cut EC2 spend by rolling out Cluster Autoscaler across three production EKS clusters and cron-based off-peak replica scaling for WebSocket workloads, while preserving headroom for latency-sensitive voice traffic
- Led the migration off a shared static AWS access key to per-workload IAM roles (IRSA) and per-instance EC2 instance profiles with least-privilege scoped policies across all clusters; manage secrets through External Secrets Operator backed by AWS Secrets Manager
- Drove end-to-end observability with cross-service Datadog APM tracing keyed on a shared correlation ID across the async call lifecycle, and stood up self-hosted Loki + Grafana across all clusters
- Architected the namespace-oriented, multi-region Kubernetes infrastructure (API, WebSocket, and self-hosted Deepgram inference clusters) with ALB routing tuned for latency-sensitive WebSocket and SIP voice services; built **kubedump**, an open-source Go CLI that snapshots live cluster state into clean, Git-trackable manifests on a daily GitHub Actions schedule
- Built a configurable LLM-based call-analysis and disposition system for per-call analytics, with model-gated execution and typed JSON configs to control cost and downstream correctness
- Stood up a new Temporal-based workflow service in production for durable, multi-step call orchestration, with an internal ALB endpoint and ExternalSecrets-backed config
- Hardened CI/CD with automated Alembic migrations and a backwards-compatibility check enforcing rolling-deploy rules, plus SHA-pinned rollbacks; stood up a dedicated webhook-delivery worker with its own pipeline

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SOFTWARE ENGINEER - PRODUCT

November 2022 - August 2025

- Engineered an LLM-powered voicebot in Go, processing millions of calls for 40+ enterprise clients.
- Rebuilt Voicebot on the open-source Pipecat-AI framework, enabling modularity and third-party integration.
- Designed and implemented a bidirectional audio-streaming telephony stack with gRPC for real-time interactions.
- Configured and maintained in-house + OpenSIPS infrastructure for call routing with telephony vendors like TCN and Twilio.
- Designed and implemented a flow-configuration validation service, completely eliminating config-related incidents.
- Maintained and enhanced the voicebot dialogue management system, adding features and improving reliability.
- Optimized flow-configuration loading, reducing memory usage by 80%.
- Served as primary on-call engineer managing incident response for critical voicebot and telephony infrastructure.
- Maintained 99.5% uptime for voicebot systems using Grafana, Loki, OpenTelemetry, Prometheus for monitoring.

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SOFTWARE ENGINEER - SOLUTIONS

December 2020 - October 2022

- Owned and maintained a critical internal repository, establishing code review processes and authoring contribution guidelines; significantly improving code quality and reducing technical debt.
- Designed and optimized voice bot flow configurations in collaboration with cross-functional teams.
- Developed 3rd-party API integrations, including PCI-DSS compliant Payment APIs, improving voice bot's debt collection capabilities
- Enhanced build pipelines, reducing deployment time and improving CI/CD efficiency.
- Maintained, customized and enhanced internal ML models used for real time conversational voice bots.
- Designed and implemented an automated daily reports system that simplified accessing and transforming on-prem data

racetrack.ai (Internship)

RESEARCH AND DEVELOPMENT TEAM

Bengaluru, India

August 2019 - March 2020

- Using data collected by automated web-scraping, published an abstract on Personalized Hybrid-Recommender Systems using social network analysis.
- Using realtime image processing, developed a minor claims automation service for car accident processing, improving claim resolution efficiency.
- Maintained and improved the performance of the primary AI chatbot product, Marvin.

ITC Infotech, Innorruption Lab (Internship)

RESEARCH AND DEVELOPMENT TEAM

Bengaluru, India

May 2019 - July 2019

- Using image processing and automation, developed a metal sheet counting algorithm.
- Using facial detection and recognition, built an ID-card detail extraction system, optimizing concierge check-in process.

Education

Institute of Technical Education and Research (ITER)

B.TECH IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Bhubaneswar, Odisha, India

2016 - 2020

- CGPA : 8.81

Ispat English Medium School (IEMS)

INTERMEDIATE/+2 FROM ISC

Rourkela, Odisha, India

2014 - 2016

- Percentage : 84.5

Ispat English Medium School

MATRICULATION FROM ICSE

Rourkela, Odisha, India

2014

- Percentage : 94.17

Skills

Languages	Python, Go, Bash, SQL
Infra & Cloud	Kubernetes (EKS), Docker, Helm, AWS (EC2, IAM/IRSA, Secrets Manager, S3, ALB, Lambda), GitHub Actions
Observability	Datadog (APM & metrics), Grafana, Loki, Prometheus, OpenTelemetry
Backend & Messaging	FastAPI, Flask, gRPC, WebSockets, RabbitMQ, Redis, PostgreSQL, SIP/OpenSIPS